

Feedback

Workshop 1

Workshop 1 feedback

- Workshop 1 was about using the Top500 list to investigate trends in HPC
- There were three sections:
 - The top 10 now and then
 - Processor trends
 - Accelerators and co-processors

Workshop 1

The top 10 now and then

The top 10 now and then

The top 10 list in Nov 2024 can be found at <https://top500.org/lists/top500/2024/11/>. The number 1 system on this list is "El Capitan". How many cores does it have and what is its LINPACK performance R_{MAX} ? What is its LINPACK efficiency (R_{MAX}/R_{PEAK})?

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	El Capitan - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TOSS, HPE DOE/NNSA/LLNL United States	11,039,616	1,742.00	2,746.38	29,581

- No. of cores = 11 039 616
- $R_{MAX} = 1742.00$ PFlop/s
- LINPACK efficiency

$$\frac{R_{MAX}}{R_{PEAK}} = \frac{1742.00}{2746.38} = 0.634 = 63.4\%$$

The top 10 now and then

The first list, published in June 1993, can be found at <https://top500.org/lists/top500/1993/06/>. The number 1 system on this list is a Thinking Machines Corporation (TMC) CM-5. How many cores did it have and what was its LINPACK performance R_{MAX} ? What was its LINPACK efficiency ($R_{\text{MAX}}/R_{\text{PEAK}}$)?

Rank	System	Cores	Rmax (GFlop/s)	Rpeak (GFlop/s)	Power (kW)
1	CM-5/1024, Thinking Machines Corporation Los Alamos National Laboratory United States	1,024	59.7	131.0	

- No. of cores = 1024
- $R_{\text{MAX}} = 59.7$ GFlop/s
- LINPACK efficiency:

$$\frac{R_{\text{MAX}}}{R_{\text{PEAK}}} = \frac{59.7}{131.0} = 0.456 = 45.6\%$$

The top 10 now and then

How do the number of cores, LINPACK performance and LINPACK efficiency compare between “El Capitan” and the fastest TMC CM-5?

	Number of cores	Linpack performance	Linpack efficiency
Thinking Machines CM5	1024	59.7 GFlop/s	0.456
El Capitan	11039616	1742.00 PFlop/s	0.634
	10780.8x		1.39X

The top 10 now and then

The 1993 top 10 includes systems which used two different vector architectures: the NEC SX-3 and Cray Y-MP systems. How do the number of cores and LINPACK efficiency of these machine compare to the TMC CM-5 and “El Capitan”?

Let's add these to the table ...

The top 10 now and then

	Rank and list	Number of cores	Linpack performance	Linpack efficiency
El Capitan	#1 November 2024	11039616	1742.00 PFlop/s	63.4%
Thinking Machines CM5	#1 June 1993	1024	59.7 GFlop/s	45.6%
NEC SX-3	#5 June 1993	4	23.2 GFlop/s	90.6%
Cray Y-MP	#9 June 1993	16	13.7 GFlop/s	90.1%

The top 10 now and then

Q: What is it about the vector architecture that gives the SX-3 and Cray Y-MP such high LINPACK efficiency?

The vector architectures used pipelining at the instruction level to give parallelism which wasn't available in other architectures at the time. The Cray and NEC architectures were also highly optimised for particular applications in contrast to the more general purpose CPUs we use today.

Pipelining is where an instruction is broken down into a series of simpler steps x . You can think of it like a production line in a factory where many cars are being assembled at once, rather than building one car then moving on the next one. Section 2.6 in the textbook has a more detailed explanation of pipelining (but this isn't something I would expect you to learn!)

The top 10 now and then

How have the top supercomputers changed between 1993 and the present?

- How do the increases in core count and LINPACK performance compare?
 - Enormous increases in LINPACK performance (>1 million times faster)
 - Processors have got faster but the achieving this increase in performance has required using many more processor cores
 - Increasing parallelism required in applications if they are to use a large fraction of a supercomputer

The top 10 now and then

How have the top supercomputers changed between 1993 and the present?

- How has the LINPACK efficiency changed?
 - Sometimes higher (NEC SX-3, Cray Y-MP) sometimes lower (CM-5)
 - Modern general purpose CPUs are acceptably good at a wide range of applications
 - Vector processors pipeline instructions: break the instruction down into smaller pieces which can be carried out independently (like a production line)
 - See [SAB sections 1.5.4 and 2.6](#) if you would like to find out more about vector processors (optional)

Workshop 1

Processor trends

Processor trends

What is the most common number of cores per socket on the current list?

What are the maximum and minimum numbers of cores per socket on the current list?

	Cores per Socket	Count	System Share (%)	Rmax (GFlops)	Rpeak (GFlops)	Cores
1	64	107	21.4	3,151,804,200	4,507,803,021	33,412,464
2	24	86	17.2	2,418,767,338	3,807,440,484	20,224,432
3	32	51	10.2	848,782,050	1,246,336,400	6,486,208
4	48	42	8.4	1,531,868,280	2,162,488,976	14,716,112
5	56	38	7.6	849,761,040	1,237,135,686	5,026,192
6	20	37	7.4	150,235,160	272,444,902	3,454,528
7	16	23	4.6	92,608,450	131,723,957	1,693,908
8	18	23	4.6	108,546,284	150,844,967	2,720,084
9	28	22	4.4	83,805,610	154,003,727	1,811,488
10	96	10	2	140,958,640	183,097,040	2,220,480
11	72	9	1.8	768,594,000	1,031,628,340	3,796,848
12	36	7	1.4	42,917,490	66,374,140	424,104
13	12	6	1.2	96,744,260	148,919,018	6,059,488
14	8	5	1	29,504,340	46,253,302	454,904
15	52	5	1	1,034,556,080	2,009,970,849	9,621,504
16	38	5	1	52,469,810	78,382,760	865,068
17	68	4	0.8	39,632,850	72,894,994	1,620,852
18	22	4	0.8	122,023,670	162,542,670	2,166,744
19	10	4	0.8	16,293,550	23,975,829	458,708
20	40	3	0.6	18,638,140	27,103,740	316,736
21	14	3	0.6	7,786,270	12,095,334	236,224
22	26	3	0.6	14,437,430	25,887,440	365,024
23	4	1	0.2	3,161,000	5,371,776	60,480
24	60	1	0.2	6,219,900	7,471,100	122,880
25	260	1	0.2	93,014,594	125,435,904	10,649,600

Processor trends

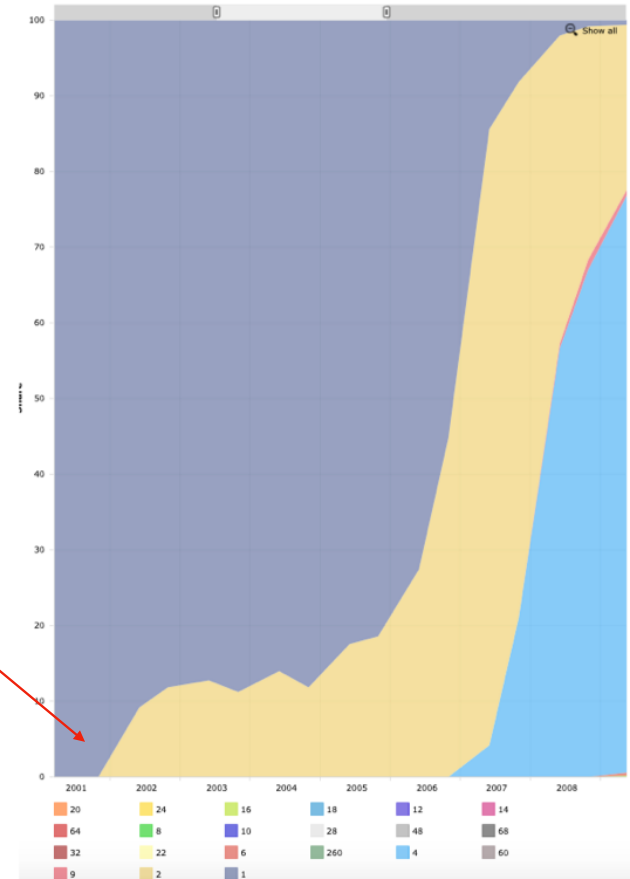
How has this changed compared to 10 years ago and 20 years ago?

	Most common	Maximum	Minimum
November 2024	>20	260	4
November 2010	4	16	1
November 2000	1	1	1

Processor trends

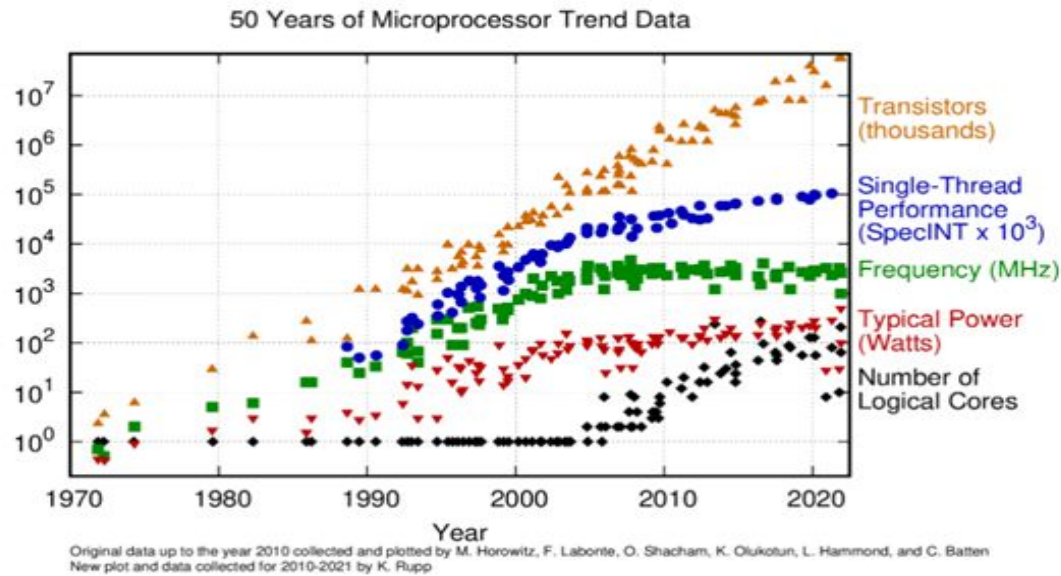
When did the number of cores per socket start increasing?

Between the November 2001 and June 2002 lists



Processor trends

Is this consistent with the trends in microprocessor design shown in figure 1?



Supercomputers get multiple cores per socket a few years earlier

Processor trends

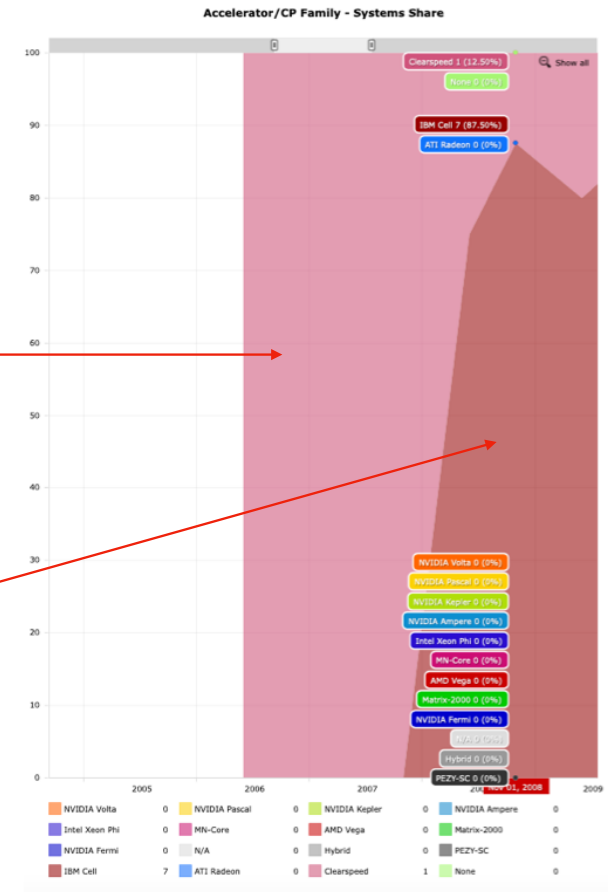
- What implications does this have for supercomputer applications if they are to make effective use of current and future supercomputers?
 - Trends in processor design continue to deliver more cores
 - Expect to need even more parallelism to make use of future supercomputers

Workshop 1

Accelerators and Co-Processors

Processor trends

- What was the first accelerator to appear on the list and when did it appear?
 - Clearspeed (June 2006): floating point accelerator card
- What was the second accelerator to appear on the list
 - IBM Cell (June 2008): as used in the Playstation 3

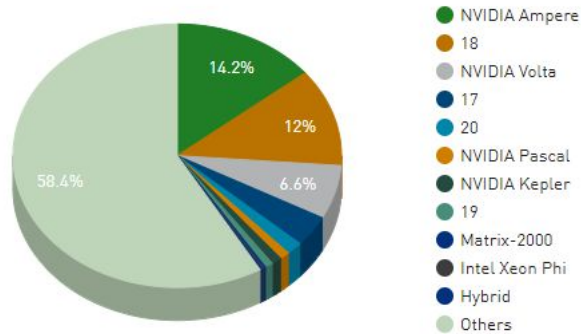


Processor trends

What type of accelerators are most popular now?
When did they start to appear?

November 2024

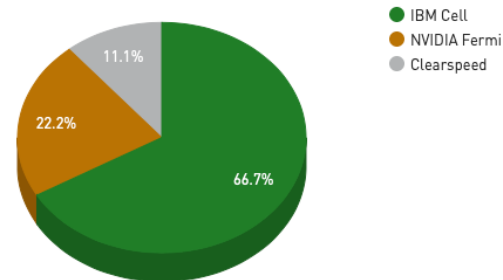
Accelerator/CP Family System Share



Mostly NVIDIA GPUs today

June 2010

Accelerator/CP Family System Share



These first appeared in the June 2010 list

Processor trends

What fraction of systems on the current list make use of accelerators?

	Accelerator/CP Family	Count ▼	System Share (%)	Rmax (GFlops)	Rpeak (GFlops)	Cores
1	Others	292	58.4	2,036,993,986	0	0
2	NVIDIA Ampere	71	14.2	1,048,725,140	1,448,388,930	9,746,476
3	18	60	12	1,967,575,660	2,965,058,760	7,080,568
4	NVIDIA Volta	33	6.6	318,290,610	484,815,792	5,894,200
5	17	19	3.8	4,444,901,070	6,632,630,582	29,086,784
6	20	8	1.6	721,794,000	958,832,260	3,574,896
7	NVIDIA Pascal	5	1	42,902,620	60,652,584	839,200
8	NVIDIA Kepler	5	1	15,412,000	25,614,070	347,984
9	19	4	0.8	1,054,038,950	2,074,398,570	9,613,760
10	Matrix-2000	1	0.2	61,444,500	100,678,664	4,981,760
11	Intel Xeon Phi	1	0.2	2,539,130	3,388,032	194,616
12	Hybrid	1	0.2	3,126,240	5,610,481	152,692

Processor trends

- How important do you think are accelerators for HPC currently?
 - A significant fraction of systems have accelerators
 - Accelerators deliver a significant fraction of the aggregate performance but they not suitable for all applications